

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25 School

of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.

5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.

4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Scenario-Based Network Design and Analysis

1. Smart Campus – Subnet Allocation

Given network: 192.168.100.0/24

Allocated subnets:

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

Subnets are assigned sequentially using VLSM.

School of Computing (/25)

- Subnet mask: **255.255.255.128**
- Block size = **128**

Parameter	Value
Network Address	192.168.100.0
First Host	192.168.100.1
Last Host	192.168.100.126
Broadcast Address	192.168.100.127
Usable Hosts	126

School of Electronics (/26)

Next available address: **192.168.100.128**

- Subnet mask: **255.255.255.192**
- Block size = **64**

Parameter	Value
Network Address	192.168.100.128
First Host	192.168.100.129

Last Host	192.168.100.190
Broadcast Address	192.168.100.191
Usable Hosts	62

School of Mechanical Engineering (/27)

Next available address: **192.168.100.192**

- Subnet mask: **255.255.255.224**
- Block size = **32**

Parameter	Value
Network Address	192.168.100.192
First Host	192.168.100.193
Last Host	192.168.100.222
Broadcast Address	192.168.100.223
Usable Hosts	30

Remaining Unused IP Range

Used range:

- 192.168.100.0 – 192.168.100.223

Unused range:

- **192.168.100.224 – 192.168.100.255**

Final Summary

Department	Subnet	Host Range	Broadcast	Usable
Computing	192.168.100.0/25	1–126	192.168.100.127	126
Electronics	192.168.100.128/26	129–190	192.168.100.191	62
Mechanical	192.168.100.192/27	193–222	192.168.100.223	30

2. Multinational Software Company

Given block: 10.10.0.0/16

Allocated subnets:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Software Development (/24)

Parameter	Value
Network	10.10.0.0
Host Range	10.10.0.1 – 10.10.0.254
Broadcast	10.10.0.255
Usable Hosts	254

Quality Assurance (/25)

Next available: 10.10.1.0

Parameter	Value
Network	10.10.1.0
Host Range	10.10.1.1 – 10.10.1.126
Broadcast	10.10.1.127
Usable Hosts	126

Human Resources (/26)

Next available: 10.10.1.128

Parameter	Value
Network	10.10.1.128

Host Range	10.10.1.129 – 10.10.1.190
Broadcast	10.10.1.191
Usable Hosts	62

Executive Management (/27)

Next available: 10.10.1.192

Parameter	Value
Network	10.10.1.192
Host Range	10.10.1.193 – 10.10.1.222
Broadcast	10.10.1.223
Usable Hosts	30

Remaining Unused Range

Used range:

- 10.10.0.0 – 10.10.1.223

Unused range:

- **10.10.1.224 – 10.10.255.255**

Final Summary

Department	Network	Host Range	Broadcast	Usable
Software Development	10.10.0.0/24	1–254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	1–126	10.10.1.127	126
Human Resources	10.10.1.128/26	129–190	10.10.1.191	62
Executive Management	10.10.1.192/27	193–222	10.10.1.223	30

3. Smart Hospital – IPv6 Migration

Given IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed Notation

Remove leading zeros and consecutive zero groups:

2001:db8:abcd::/64

2. Expanded Form

Expanded back to full hexadecimal:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix

With /64, the first 64 bits represent the network.

Network Prefix:

- Full: 2001:0db8:abcd:0000::/64
- Compressed: **2001:db8:abcd:0::/64**

4. Host Addresses Supported

Host bits:

- $128 - 64 = 64$

Number of addresses:

$2^{64} = 18,446,744,073,709,551,616$

Final IPv6 Summary

Task	Result
Compressed	2001:db8:abcd::/64
Expanded	2001:0db8:abcd:0000:0000:0000:0000:0000

Task	Result
Network Prefix	2001:db8:abcd:0::/64
Host Bits	64
Total Possible Hosts	18,446,744,073,709,551,616

This enormous address space is ideal for **IoT-enabled smart hospitals** with thousands of connected devices.

4. ISP Routing Scenario

Routing Table

Destination Network
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Incoming packet destination:

192.168.2.45

1. Destination Subnet

Subnet mask /24 = 255.255.255.0

Network portion:

- 192.168.2.45 → 192.168.2.0

Destination subnet: 192.168.2.0/24

2. Matching Routing Entry

The router searches for the **longest prefix match**.

Matching entry:

192.168.2.0/24

3. Next-Hop / Interface

The packet is forwarded through the interface connected to:

Network 192.168.2.0/24

Example:

- **Interface:** GigabitEthernet0/1
- **Next-hop network:** 192.168.2.0/24

4. Default Route

If no route matches, the router should use:

0.0.0.0/0

Example:

0.0.0.0/0 → ISP Gateway

Routing Summary

Item	Result
Destination IP	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Forwarding Interface	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0

5. Layer-3 Router Interconnecting Two LANs

Given

- **Administration LAN:** 192.168.10.0/26
- **Finance LAN:** 192.168.10.64/26

Router interfaces:

- Administration Gateway → 192.168.10.1

- Finance Gateway → 192.168.10.65

A /26 subnet provides:

- 64 total addresses
- 62 usable hosts

Administration LAN

Network Details

Parameter	Value
Network Address	192.168.10.0
First Host	192.168.10.1
Last Host	192.168.10.62
Broadcast Address	192.168.10.63

Suggested Computer IPs

Computer	IP Address
PC-A1	192.168.10.10
PC-A2	192.168.10.11
PC-A3	192.168.10.12

Default Gateway

192.168.10.1

Finance LAN

Network Details

Parameter	Value
Network Address	192.168.10.64
First Host	192.168.10.65
Last Host	192.168.10.126
Broadcast Address	192.168.10.127

Suggested Computer IPs

Computer	IP Address
PC-F1	192.168.10.70
PC-F2	192.168.10.71
PC-F3	192.168.10.72

Default Gateway

192.168.10.65

Final LAN Summary

LAN	Network	Host Range	Broadcast	Gateway
Administration	192.168.10.0/26	1–62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	65–126	192.168.10.127	192.168.10.65

Overall Interpretation

These scenario-based solutions demonstrate practical application of:

- **IPv4 subnetting using VLSM**
- **Network, broadcast, and host range identification**
- **Enterprise IP address planning**
- **IPv6 compression and prefix analysis**
- **Routing table lookup and longest-prefix matching**
- **Layer-3 inter-network communication using default gateways**

Such structured network design ensures **efficient address utilization, scalability, security segmentation, and reliable communication across campus, enterprise, ISP, and smart healthcare environments.**